

WORK EXPERIENCE

Business Intelligence Engineer

June 2025 – Present

Amazon, Austin, Texas

- Led the redesign of a large-scale A/B experiment after identifying critical outlier concentration (top 1% products drove 75% of effect). Implemented statistical controls to ensure reliable causal estimates and saved \$500K+ in experimental costs.
- Developed causal ML model using Causal Forest and doubly-robust estimators to decompose aggregate treatment effects into individual product-level impacts, scaling to 10M+ products with propensity-weighted causal validation.
- Architected and maintained distributed ETL pipelines processing terabytes of daily data from multiple sources. Built real-time executive dashboards tracking critical KPIs and product launch metrics for Amazon's defect elimination program.
- Developed AI-powered analytics agent fine-tuned on internal schemas and queries, enabling natural language data exploration with autonomous SQL and Python generation, execution, and visualization via MCP server architecture.

Machine Learning Engineer

September 2024 – June 2025

Prompt Inversion AI, Dover, Delaware

- Architected multi-agent spam classifier benchmarking 6 agentic frameworks (LangGraph, PydanticAI, AutoGen, CrewAI, Swarm, SmolAgents) with RLHF pipeline, achieving 97% accuracy on 500K+ daily messages.
- Built LLM red-teaming platform with 10+ jailbreak algorithms and a quantitative eval framework, testing DeepSeek-R1/V3 across 5000+ adversarial scenarios to establish production safety baselines.
- Redesigned FastAPI backend with async architecture, implementing Redis caching and connection pooling to handle 100K+ requests/day, cutting response time by 60% for workflows processing 2M+ monthly interactions.
- Engineered multi-stage candidate ranking algorithm combining vector embedding retrieval, LLM semantic scoring, and keyword matching to achieve 40% improvement in accuracy over baseline methods across 10K+ resumes.

Analytical Scientist Intern

June 2023 – December 2023

FICO, San Diego, California

- Developed automated drift monitoring system for production fraud detection models using a custom time-series algorithm, tracking distribution shifts across 15 enterprise clients, and reduced false positive alerts by 90%.
- Productionized unsupervised clustering pipeline (tSNE) to identify emerging fraud patterns, enabling proactive model retraining.

AI Engineer

February 2022 – August 2022

Collablens, Haryana, India

- Engineered production-grade computer vision systems for automated manufacturing quality control: integrated an edge-cloud spillage detection pipeline with real-time video streaming, and a laser-engraving defect classifier achieving 95% accuracy with 2.5s inference, enabling automated assembly line quality control.

EDUCATION

University of California San Diego

2022 – 2024

Master of Science, Data Science

GPA (3.98/4.0)

Delhi Technological University, New Delhi

2018 – 2022

Bachelor of Technology, Mathematics and Computing Engineering

CGPA (8.73/10)

PROJECTS

[Rubik's Cube 3D Visualizer and Deep Reinforcement Learning Solver](#)

July 2024 – September 2024

- Implemented deep RL solver combining MCTS with custom ResNet policy/value networks trained via self-play on 10M+ game states. Achieved 71% solve rate on 9-move scrambles with <1s inference, demonstrating efficient search space pruning.

[MediLoRA: LLM for medical Q&A with QLoRA](#)

October 2023 – January 2024

- Fine-tuned 7B medical LLM using QLoRA on 300M tokens (PubMed, medical textbooks), achieving 20+ point accuracy gains on PubMedQA/MedQA. Matched 70B SOTA performance on MMLU-Medical using 0.05% of training compute.

RESEARCH EXPERIENCE AND PUBLICATIONS

- [Sood, S., Jain, T., Batra, N., Taneja, H.C. \(2023\). Black-Scholes Option Pricing Using Machine Learning.](#) [ICDSA 2023]
- [S. Sood and Y. Ahuja, "Selective Lossy Image Compression for Autonomous Systems."](#) [STSIVA 2021]

TECHNICAL SKILLS

- **Programming :** Python, SQL, JavaScript, C++, R, MATLAB, HTML, CSS
- **Technologies :** Pandas, NumPy, PyTorch, TensorFlow, Git, REST, AWS, PostgreSQL, Spark, Docker, React, Next.js
- **Skills :** Data Science, Machine Learning, Data Engineering, Causal Inference, Reinforcement Learning, System Design